LEAP Program Funding Analysis

A Five-Phase Quantitative Framework

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Roadmap

Problem & Framework

- What is LEAP?
- Why is this hard?
- Our five-phase approach

The Model

- Phase 1 — Deterministic baseline
- Phase 2 — Monte Carlo simulation
- Phase 3 — Property-level
- Phase 4 — Competing risks
- Phase 5 — Final answer

Key Findings

- The repayment problem
- What drives the cost
- Loan vs. grant
- Refinancing surprise

Recommendations

- Final funding tiers
- Policy actions

The LEAP Program: A Unique Financing Challenge

What is LEAP?

Columbus offers homeowners a **zero-interest, deferred-repayment loan** (up to $10,000) to replace lead or galvanized private-side water lines, mandated federally by 2037.

Loan Structure

- 0% interest, up to 99 years
- Repaid as a *single lump sum*
- Triggered by: sale, refinancing, or HELOC
- If inherited and not sold: no repayment required

Why This Is Hard

- City deploys $100M+ over 12 years
- No predictable return schedule
- Repayments driven by housing markets and Fed policy
- Must stay solvent through 2037 with no guarantees

Our Objectives

The Central Question

How much must Columbus set aside **in 2026** to keep LEAP solvent through 2037?

1. **Estimate total upfront funding** required (2026–2037)
2. **Determine a funding level sufficient with 95% probability** under our model
3. **Model realistic uncertainty:** participation rates, breakage, repayment timing, interest rates, and inheritance
4. **Compare loan vs. grant** structure on true economic cost

Five Phases, Each Adding a Layer of Realism

Phase	Type	What It Adds
1	Deterministic	Best-estimate baseline, sensitivity grid
2	Monte Carlo	Stochastic uncertainty, breakage scenarios
3	Micro-simulation	Property heterogeneity, non-repayment
4	Competing Risks	Refi/HELOC channels, interest rate paths
5	Meta-Distribution	True 95% CI across all scenarios

Data: Pilot funnel · 63 quotes · WSL insurance · 2024 CFPB HMDA (43K records) · Census ACS B25026 · SIAM-MTI property dataset

How Many Properties Are Eligible? (24,416)

Customer-side material is **unknown** for 59,973 properties. We use a **Bayesian conditional probability** approach:

$$\hat{p}(u) = \frac{\text{known hazardous with utility } u}{\text{known total with utility } u}$$

Apply $\hat{p}(u)$ to each unknown-customer group:

Utility Material	Hazard Rate
Lead / Galvanized	33%
Non-Lead Copper	10%
Plastic / Other	<0.2%
Unknown	17%

Result

14,354 confirmed
+ 10,062 estimated

24,416 eligible

Phase 2 models this estimate as $\mathcal{N}(24{,}416,\ 1{,}006^2)$ — 10% uncertainty on the estimated component.

Phase 1: Baseline Requires $113.5M — But It's Not Enough

Fixed inputs:

- Participation: 51.22% (42/82 pilot)
- Avg. loan: $7,582 · +3%/yr
- Repayment lag: 12 years
- Mandate: 2%–10%/yr through 2037

Loans issued: ∼12,506

The Repayment Problem

Under a 12-year lag, **zero dollars return before 2038.** The program is a grant during its own window.

Phase 1 Result

$113.5M

deterministic baseline

Key sensitivity: Participation rate dominates — 33% → 89% uptake changes capital by **2.7**×. Beyond 12 years, repayment lag has **zero** impact on near-term capital needs.

Phase 2: Uncertainty Raises the Bar to $165M+

Replace fixed assumptions with distributions. Run 10,000 simulations.
Stochastic inputs:

- Universe: $\mathcal{N}(24{,}416,\ 1{,}006^2)$
- Participation: Beta(42, 40) — calibrated directly to pilot: 42 successes, 40 failures
- Repayment lag: LogNormal(mean=12, std=5)

Three breakage scenarios:

Scenario	Rate
Primary (pilot)	0.24%
Adjusted	**1.38%**
Adverse (insurance)	3.44%

95th Percentile Results

Primary	$134.0M
Adjusted	**$164.7M**
Adverse	$218.2M

Critical Finding

$113.5M is **insufficient 40.7%** of the time. The Phase 1 estimate is not a conservative number.

The Breakage Rate: A 14× Discrepancy

Two independent sources disagree sharply:

Source	Rate	Issue
Pilot data	0.24%	Denominator too large
Insurance data	3.44%	Selection bias
Adjusted	**1.38%**	**Corrected**

Insurance policyholders *self-select* into coverage because they suspect risk. Their 3.44% rate overstates the general population. We apply a **2.5× selection bias correction**: $1.38\% = 3.44\% \div 2.5$.

Why It Matters

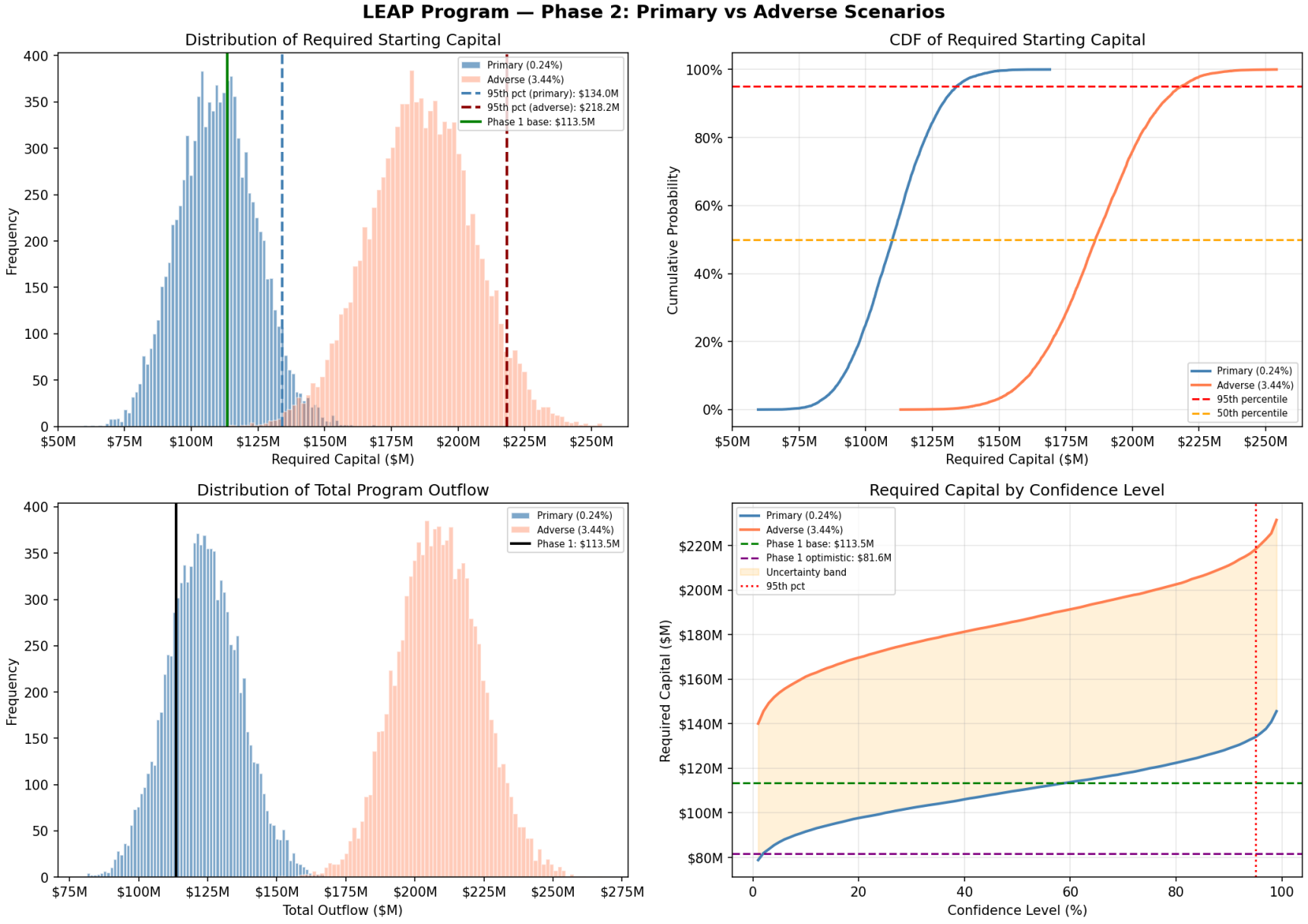
The primary → adjusted gap is **$30.7M** at the 95th percentile.

Not knowing the true rate costs $37.5M in required reserves.

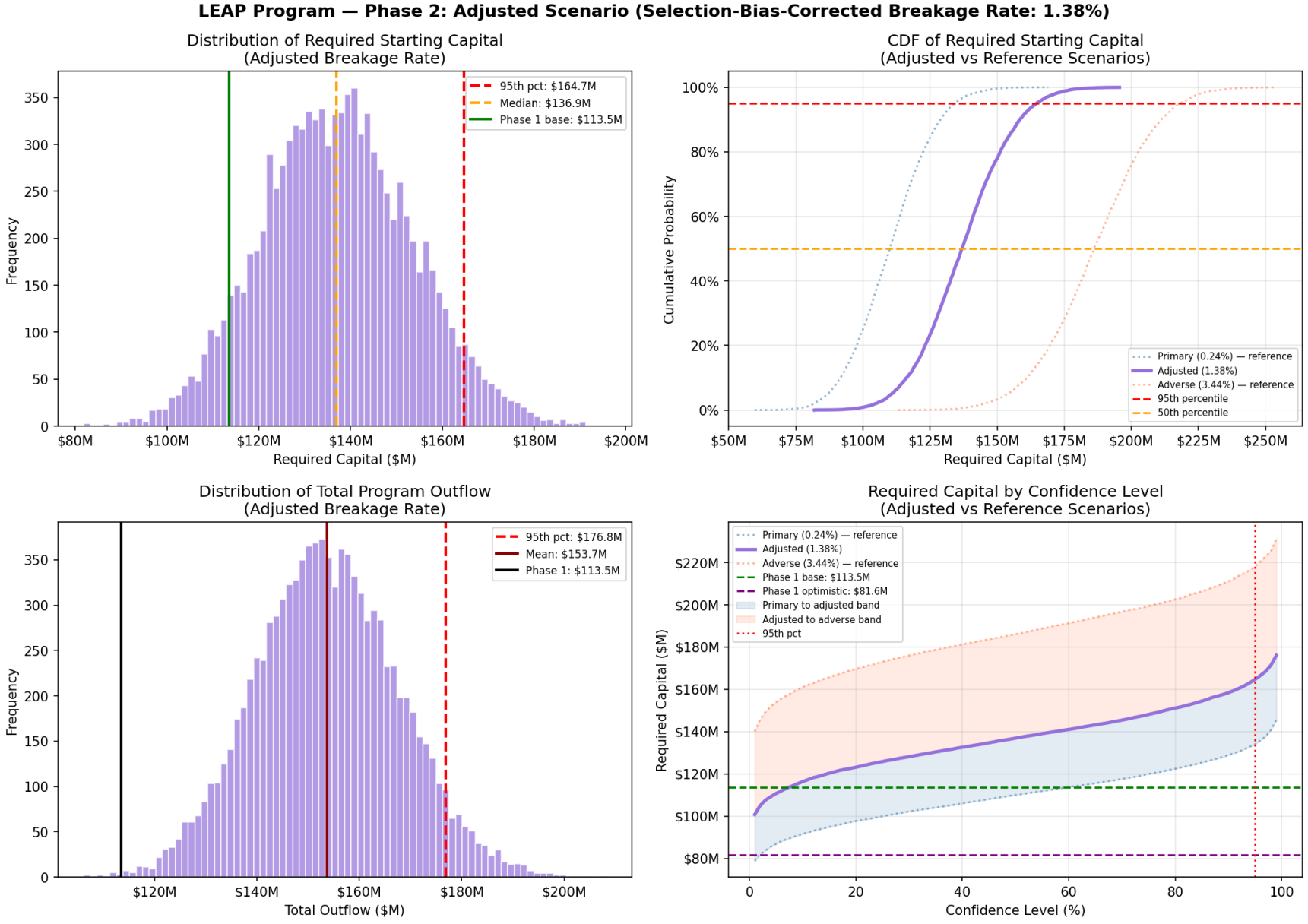
The Fix

Two more years of pilot breakage data collection could resolve most of this uncertainty.

Phase 2: Primary vs. Adverse Scenarios



Phase 2: The Adjusted Scenario Is Our Recommendation



Phase 3: $13M Will Never Be Repaid

What Phase 3 adds:

- Models 6,057 properties individually
- Logistic regression: participation by property type, occupancy, assessed value, ownership length
- Property-type repayment lags (8–14 yr means)
- Permanent non-repayment via inheritance

Type	Tenure	Non-Repay
Single family	Owner	10%
Duplex	Owner	15%
Multi-family	Owner	20%
Any	Investor	8–12%
Blended		**11.57%**

The Finding

$13.0M

mean permanent loss

12 cents per dollar loaned

Key Insight

The law of large numbers dominates at 6,057 properties — heterogeneity does not increase aggregate uncertainty. Phase 3's contribution is *quantifying the permanent loss.*

Phase 4: Refinancing Rivals Property Sales

Three explicit competing triggers, calibrated to Columbus data.
Data sources:

- **Census B25026:** Weibull sale hazard from Columbus tenure distribution (k=1.17, λ=13.9 yr)
- **CFPB HMDA:** Refi rate 2.33%, HELOC 0.75% at 7.44% avg rate (43,276 Columbus records)
- **Vasicek model:** Stochastic rate paths, three macro scenarios

Trigger	Share
Property sale	38.1%
Refinancing	**38.7%**
HELOC	8.6%
Never (non-repay)	14.6%

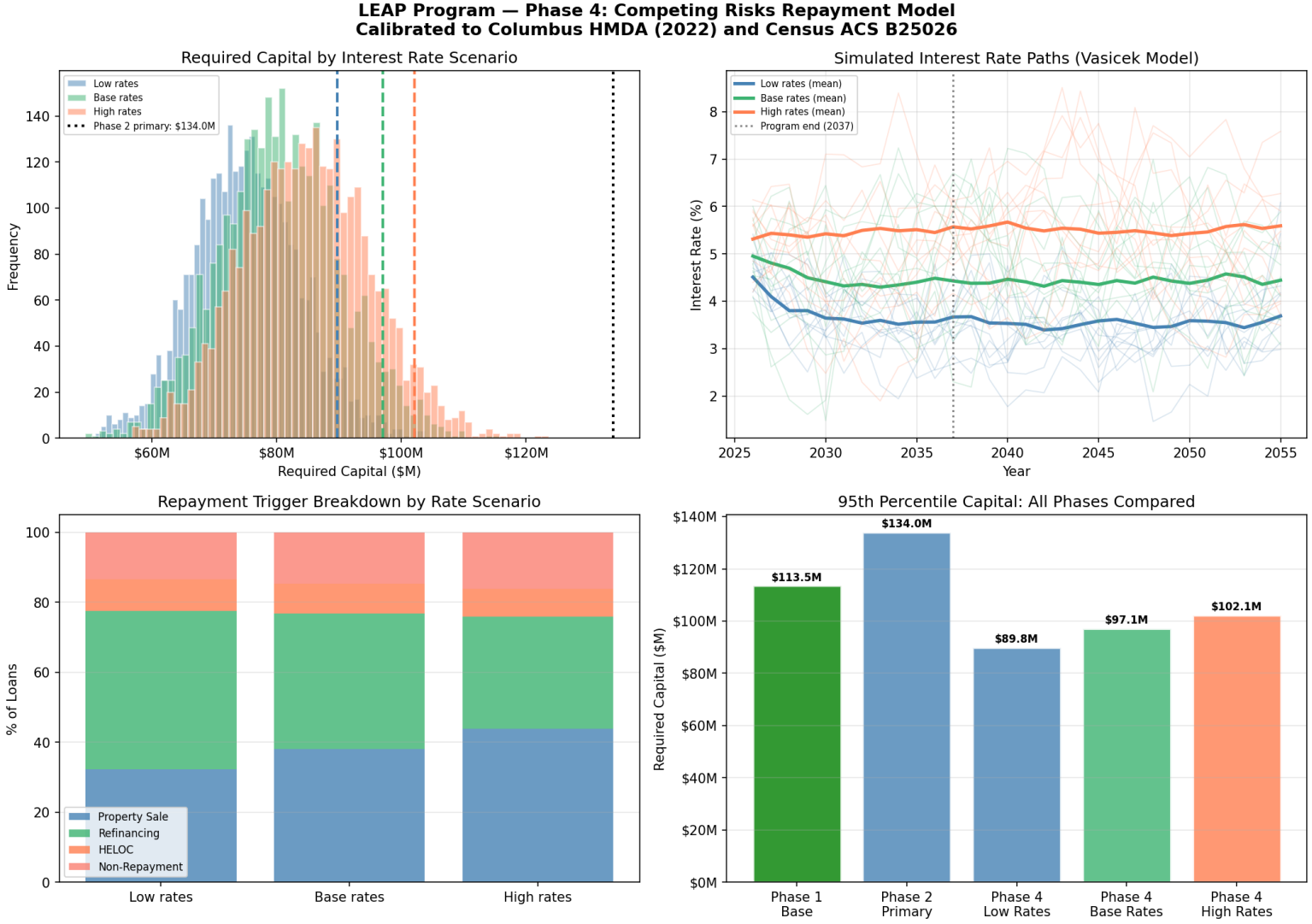
Surprise Finding

Refi accounts for **38.7%** of repayments — nearly matching property sales.

Fed rate policy directly affects LEAP cash flows. A rate cut cycle reduces required capital by \$30–44M.

2-year seasoning applied before refi/HELOC channels activate — homeowners do not immediately refinance after a home improvement loan.

Phase 4: Interest Rates Create a $12M Funding Range



95th pct: *Low rates $89.8M · Base $97.1M · High $102.1M* · **Rate sensitivity: $12.4M**

Phase 4 is lower than Phase 2 because refi/HELOC channels bring repayments into the program window.

Phase 2 remains the conservative planning figure if equity extraction is uncertain.

Phase 5: A True 95% CI Across All Scenarios

We do not know which breakage scenario is correct — so we weight all three.

Scenario	Weight	Rationale
Primary (0.24%)	20%	Likely understated
Adjusted (1.38%)	**55%**	**Best estimate**
Adverse (3.44%)	25%	Correction uncertain

100,000 samples drawn in proportion to weights $\rightarrow$ single unified distribution.

Results

Median: \$139.1M
95th pct: \$202.5M
95% CI: [\$93M, \$211M]

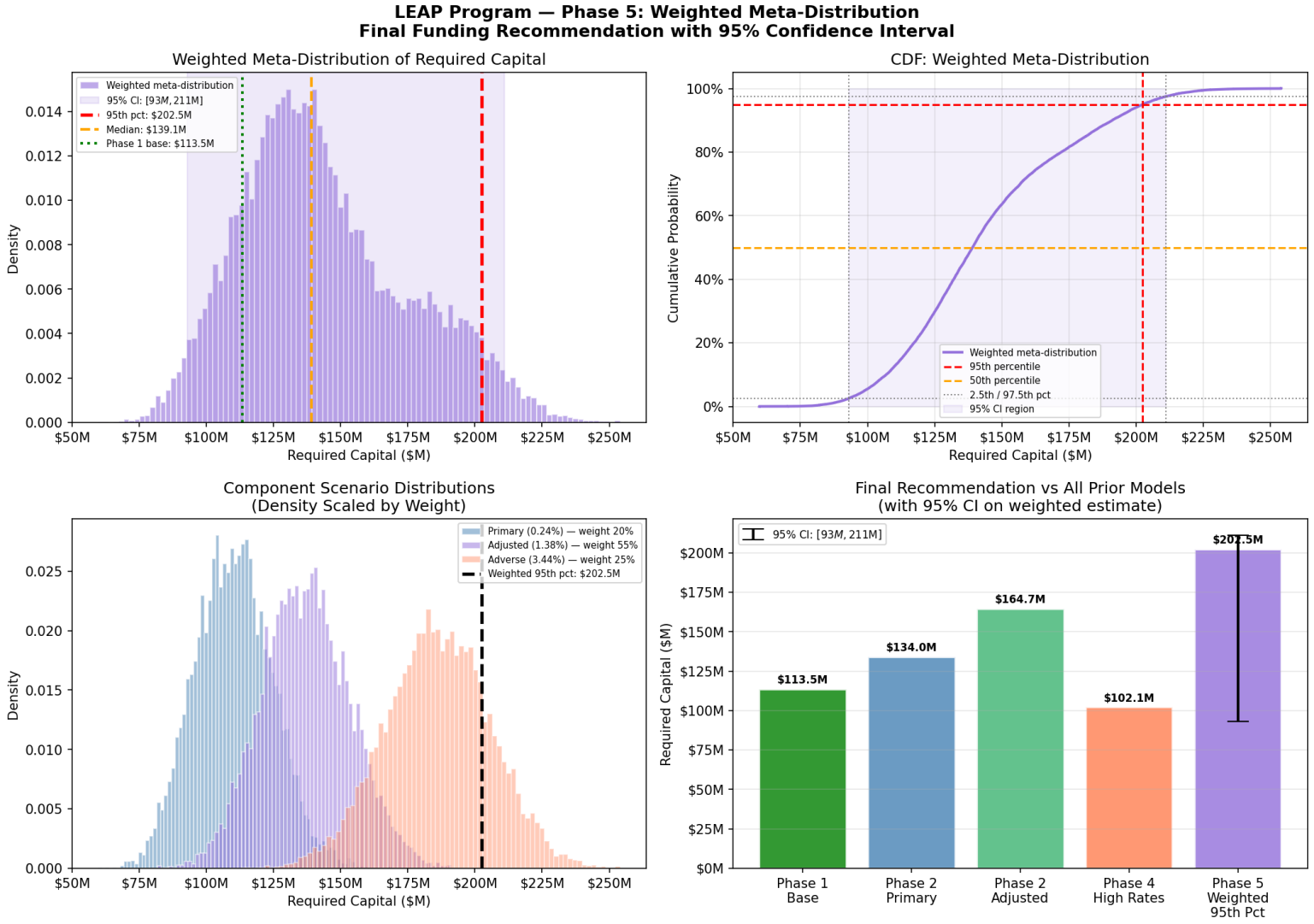
Why \$202M > \$165M?

Mixing distributions with means of \$110M, \$137M, \$186M inflates the upper tail.

The **\$37.5M gap** is the cost of *not knowing* the true breakage rate.

No defensible weighting produces a number below \$165M.

Phase 5: Weighted Meta-Distribution



The 95th percentile of $202.5M (red dashed) and the 95% CI (shaded region) capture both within-scenario simulation uncertainty and between-scenario model uncertainty simultaneously.

Methods: Chosen for a Reason, Not for Complexity

Method	**Why We Used It**
Beta(42, 40) distribution	Encodes pilot exactly: 42 completions, 40 non-completions
Log-normal distribution	Right-skewed; correct for positive costs and lags
Monte Carlo simulation	Propagates multiple uncertainties simultaneously
Logistic regression	Predicts participation from observable property features
Weibull hazard model	Sale probability increases with homeowner tenure
Vasicek model	Captures mean-reverting nature of interest rates
Competing risks	Models three independent repayment triggers at once
Mixture distribution	Formally combines uncertainty *across* scenarios

Key Assumptions Across All Phases

Participation & Universe

- Rate: 51.22% — Beta(42, 40)
- Universe: 24,416 — Normal(σ=1,006)
- Cost: \$7,582 + 3%/yr escalation

Repayment (Phase 2)

- Lag: LogNormal(mean=12, std=5 yr)
- Breakage: 0.24% / 1.38% / 3.44%

Non-Repayment (Phase 3)

- Blended: 11.57% (8–20% by type)

Competing Risks (Phase 4)

- Sale: Weibull(k=1.17, λ=13.9 yr)
- Refi: 2.33%/yr at 7.44% (HMDA)
- HELOC: 0.75%/yr at 7.44% (HMDA)
- Rates: Vasicek($\bar{r}$=3.5–5.5%)
- Seasoning: 2 yr before refi/HELOC
- Selection effect: 40% sale hazard for first 5 yr post-loan

Economic Cost:

Opportunity cost 3.5%/yr on reserves · LEAP uses operating funds, not bonds

Loans Are Unambiguously Superior to Grants

Model	Upfront	Long-Run Net	True Cost
Grant	\$113.5M	\$113.5M	\$113.5M
Loan (L=12)	\$113.5M	\$0	\$47.7M
Loan (L=7)	\$81.6M	\$0	\$27.8M

Loan true economic cost:

- Financing subsidy: \$47.7M
- Non-repayment loss: \$13.0M
- **Total: ≈\$60.7M**

The Verdict

Loans cost **53%** of an equivalent grant. The city recovers principal over 20–30 years — paying only **42 cents per dollar** in true economic terms.

Six Findings That Shape the Recommendation

1. **Phase 1's $113.5M fails 40.7% of the time.** A deterministic estimate is not a conservative estimate.
2. **Participation rate dominates everything.** 33% $\rightarrow$ 89% uptake changes required capital by 2.7$\times$.
3. **Breakage rate uncertainty costs $37.5M.** The gap between $165M and $202.5M is entirely explained by not knowing the true pipe failure rate.
4. **Refinancing rivals property sales as a trigger.** 38.7% of loans repay via refi — never captured before Phase 4. Fed rate decisions directly affect LEAP cash flows.
5. **$13M will never be repaid.** Inheritance creates a permanent loss that must be budgeted.
6. **Beyond 12 years, repayment lag is irrelevant.** All such loans repay post-2037 regardless. The 12–20 year range has zero impact on near-term capital needs.

All Results at a Glance

Phase	Key Finding	95th Pct
Phase 1 (Deterministic)	Point estimate; insufficient 40% of the time	$113.5M
Phase 2 Primary	Breakage likely understated; lower bound	$134.0M
Phase 2 Adjusted	**Best single estimate; recommended**	**$164.7M**
Phase 2 Adverse	Stress test; no selection bias correction	$218.2M
Phase 3	$13M permanent non-repayment identified	$102.0M*
Phase 4 High Rate	Refi/HELOC active; conditional lower bound	$102.1M
Phase 5 Weighted	True 95% CI across all scenarios	$202.5M

* Phase 3 models voluntary channel only; not directly comparable to Phase 2

The Answer: A Tiered Funding Framework

Tier	Amount	Meaning
Conditional floor	\$102M	Refi/HELOC active (Phase 4)
Absolute floor	\$113.5M	Phase 1; 40% failure rate
Lower bound	\$134M	Breakage understated
Recommended	**\$165M**	**95% confidence, best estimate**
Risk-averse	\$202.5M	95% incl. model uncertainty
Stress test	\$218M	Worst credible scenario

95% Confidence Interval (Phase 5)

[\$93M, \$211M] point estimate: **\$202.5M**

Fund \$165M if you trust the adjusted breakage rate. Fund \$202.5M to hedge against model uncertainty. The \$37.5M gap is the price of not knowing the true failure rate.

Financial Feasibility: $165M Is Manageable

It deploys over 12 years, not all at once:

- 2026–2028: $2–8M/yr (ramp-up)
- Peak: ∼$12.5M/yr in 2036
- Full $165M never needed liquid simultaneously

True economic cost is $60.7M — not $165M:

- Most capital is recovered over 20–30 years
- ≈$2M/yr in true cost — well within Columbus's $1.1B+ operating budget
- Federal EPA Dwsrf grants can offset further

Federal Support Available

EPA DWSRF: Drinking Water State Revolving Fund provides low/zero-interest financing for exactly this type of program.[1]

BIL: The Bipartisan Infrastructure Law allocated **$15B nationally** for lead service line replacement.[2]

LCR Improvements: EPA's 2024 Lead and Copper Rule requires replacement by 2037 and anticipates federal cost-sharing.[3]

Policy Recommendations: Capital and Structure

Capital Reserve

- **Fund $165M** — baseline, 95% confidence
- **Consider $202.5M** if risk-averse
- $134M is the absolute floor
- $218M is the stress test threshold

Program Structure

- **Loans, not grants**
  Save $53M in true economic cost
- **Budget $13M** non-repayment reserve
- **Prioritize outreach**

Why Loans Win

Same upfront cost as a grant in the base case — but the city recovers its capital and pays only 42 cents per dollar in long-run economic cost.

Most Impactful Program Action

Invest in participation outreach. Moving from 33% to 51% participation changes required capital by $40M. No other lever comes close.

Policy Recommendations: Monitor and Collect Data

Monitor Two Leading Indicators Annually

Housing Turnover

Primary repayment trigger. A market slowdown extends the deployment period and delays capital recovery.

Federal Reserve Rate Policy

Governs the refi and HELOC channels. A rate cut cycle could reduce required capital by $30–44M relative to Phase 2.

Highest-Value Investment

Collect two more years of systematic breakage data.

The $37.5M gap between $165M and $202.5M is driven entirely by breakage rate uncertainty. Better data reduces this gap — potentially saving tens of millions in required reserves.

Also worth collecting: County probate records (empirical non-repayment rates) and individual deed transfer data (Weibull calibration).

Limitations and Extensions

Current Limitations

- Property dataset is simulated (Phase 3)
- Weibull sale hazard overestimates by $\sim$3 pp due to grouped Census data
- HMDA anchored to 2022 high-rate environment
- Refi and HELOC treated as independent events
- Phase 2 assumes 100% eventual repayment

Natural Extensions

- Real property assessment records (Phase 3)
- Individual deed transfers for Weibull calibration
- County probate records for non-repayment rates
- Correlated inputs: participation $\leftrightarrow$ turnover
- Unified Phase 4+5 model combining rate and breakage uncertainty simultaneously

Conclusion

What We Built

A five-phase quantitative framework using real program data, CFPB HMDA records, and Census ACS data to answer a municipal finance problem.

Fund $165M
95% confidence under best breakage estimate

Consider $202.5M
95% confidence including model uncertainty

True economic cost: $60.7M
vs. $113.5M grant equivalent

Loans win.
Always, by $53M in long-run cost.

Watch participation rate
Dominates every other variable

Watch the Fed
Rate cuts save $30–44M in required capital

Budget $13M for non-repayment
Real, quantified, unavoidable

Collect breakage data
Worth $37.5M in reduced uncertainty